SWAPNIL MITRA

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Software engineer specialising in cloud integration, backend systems, and AI-enabled tooling. I build dependable platforms that connect complex systems and take them from architecture through to production.

TECHNICAL SKILLS

Languages
C#, TypeScript, JavaScript, Python, Rust, SQL, Solidity

Frameworks
.NET 8, ASP.NET Core, React, Next.js, Flutter, Express, FastAPI, Fluent UI, Tailwind CSS

AI & Data
Claude API, OpenRouter, LLM tool calling, LangChain, AWS Bedrock, RAG pipelines, pgvector, SQL Server, PostgreSQL, MongoDB, Prisma, Drizzle

Azure
Functions, API Management, Logic Apps, App Service, Entra ID, Key Vault, Application Insights, Monitor

DevOps & Messaging
GitHub Actions, Azure DevOps, Docker, RabbitMQ, Prometheus, Grafana

Platforms & Tools
AWS (S3, Transcribe, Polly), Microsoft Graph, SharePoint, OAuth2, SOAP, Hardhat, IPFS, Git

PROFESSIONAL EXPERIENCE

Adrosonic | **Research and Innovation Engineer** · *Jul 2023 – Present*

Insurance Broker Management Platform

- Led a team of 4 on a broker management platform (.NET, SQL Server, MongoDB on Azure), used daily by insurance brokers across Australia and New Zealand to quote, bind, amend, and cancel policies.
- Delivered Microsoft single sign-on and transactional policy updates, so no policy change is ever left half-processed.
- Architected the asynchronous integration layer connecting the platform to insurers on Azure API Management, Logic Apps, and Functions, cutting integration time from ∼6 weeks to ∼2 with 99.9% successful delivery.
- Added end-to-end tracking and alerting with Application Insights and Azure Monitor; wrote the handover docs.

Policy Administration Integration

- Led delivery of a single API on Azure Functions giving external callers one interface to several policy administration systems, routing each request to the right system, merging results, and handling slow writes asynchronously.
- Built on stateless per-system adapters behind one shared contract, so new consumers onboard without API changes.

Release Engineering

- Led the team that built a shared, versioned CI/CD framework on GitHub Actions that any repository adopts with minimal config, standardising build, test, and release for .NET, Azure Functions, NuGet, and Logic Apps.
- Added release safeguards: pull-request quality gates linked to Azure DevOps work items, approval-gated Staging and Prod releases with fast slot-based rollback, OIDC-based Azure access, and Key Vault-only secrets.

Email Filing & Document Management

- Rebuilt an email filing and SharePoint document pipeline on .NET and RabbitMQ publish/subscribe messaging with duplicate protection, dead-letter queues, and Prometheus monitoring, cutting message latency ∼40%.
- Replaced a legacy Outlook add-in with a React and Fluent UI web add-in with Microsoft sign-in, Graph integration, and offline caching, rolled out to 10,000+ users.

PROJECTS

- **Heptapod** · *Sep – Oct 2026* Lets one LLM agent operate enterprise systems with different protocols (REST, SOAP, SQL) and auth through one tool interface; new systems plug in without agent changes (Python).
- **MemVault** · *Aug 2026* Local-first memory engine for AI agents in Rust with a tamper-evident record of everything stored and retrieved, provable deletion, hybrid semantic and keyword search, and MCP and Python bindings.
- **Groundwork** · *2024 – 2026* AI outreach workspace that reads a job posting and drafts evidence-backed emails with an LLM over hybrid search, gated by human approval and an audit trail (Next.js, Postgres + pgvector).
- **AGRISETU** · *Feb – Apr 2026* Voice-first AI marketplace (Flutter, Next.js, Express) where farmers order supplies together by voice in 11 languages, built on AWS Transcribe, Polly, a Bedrock LLM, and a RAG knowledge base.

EDUCATION

Indian Institute of Technology, Kharagpur | **B.Tech, Biotechnology and Biochemical Engineering** · *May 2023*

EXTRA-CURRICULARS

- Associate Manager, E-Cell IIT KGP: organised Global Entrepreneurship Summit 2021, LSM, EAD, Empresario 2020.
- National Service Scheme Volunteer, IIT KGP: Silver Medal for collaborative performance (2019).
- Club Service Director, Rotaract Club of Calcutta North East.